

9 Adenosine triphosphate (ATP) is used by organisms as an immediate source of energy.

(a) List **two** anabolic reactions in cells that need ATP.

.....

..... [2]

(b) Figure 9.1 shows three stages in the life cycle of a marine invertebrate, the jellyfish *Aurelia aurita*.

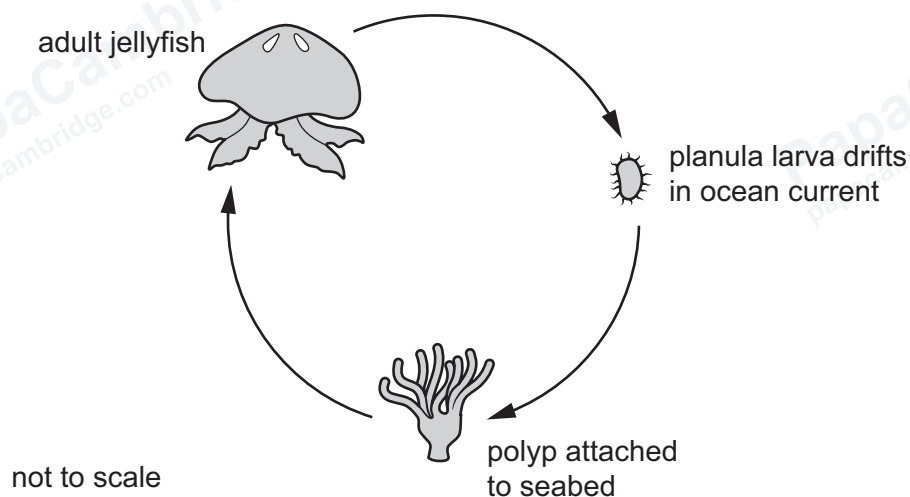


Figure 9.1

(i) Suggest why the adult jellyfish needs more ATP than the other life cycle stages.

.....

.....

.....

..... [2]

(ii) State **two** ways in which ATP can be synthesised in cells, such as those in *Aurelia aurita*.

.....

.....

.....

..... [2]

- (c) The respiratory quotient (RQ) can identify the respiratory substrate that is being used to synthesise ATP in an organism. A respirometer can be used to calculate the RQ.

Table 9.1 shows the volumes of oxygen used and carbon dioxide produced when three different types of substrate are respired.

Table 9.1

respiratory substrate	volume of oxygen used / cm ³	volume of carbon dioxide produced / cm ³
A	830	830
B	2020	1430
C	970	780

- (i) Use Table 9.1 to calculate the RQ for respiratory substrate **B**.

Show your working and give your answer to **two** significant figures.

RQ = [2]

- (ii) With reference to Table 9.1, state the letter of the respiratory substrate that is a carbohydrate such as glucose.

..... [1]

[Total: 9]